GUARDIANSIM

GuardianSim: Counterfactual Safety Certification for Robot Manipulation on AMD Radeon GPUs

Track 3 - Physical AI Challenge
AMD Radeon Cloud / ROCm / Genesis simulation

REVIEW DRAFT - Luma rules sign-off and final video pending

Project repository
github.com/nvm-star-max/GuardianSim

Team attribution
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Counterfactual safety certification for robot manipulation on AMD Radeon GPUs

Abstract

Robot manipulation policies are usually optimized to complete a task, but a nominally successful grasp can still pass too close to nearby objects, contact clutter, or become unstable when scene geometry changes. GuardianSim is a safety-oriented decision layer for simulated Franka Panda fruit picking. Given a nominal grasp, it generates counterfactual alternatives, restores every alternative to the same Genesis scene snapshot, measures physical rollout outcomes on an AMD Radeon GPU, and selects an action only when it satisfies frozen reachability, stability, clearance, and repeatability requirements. If no candidate passes every hard gate, the system explicitly stops.

In the primary frozen Gate 3.2 benchmark, GuardianSim achieved repeatable safe completion in 30 of 30 unseen clutter scenarios, compared with 18 of 30 for the nominal baseline. Across three independent physical executions per scenario, safe executions improved from 58/90 to 90/90, while sampled clutter contacts decreased from 30 to zero. Mean sampled clutter clearance increased from 0.023191 m to 0.046003 m. These are Genesis simulation results on AMD Radeon Cloud, not physical-robot deployment claims.

1. Target application and motivation

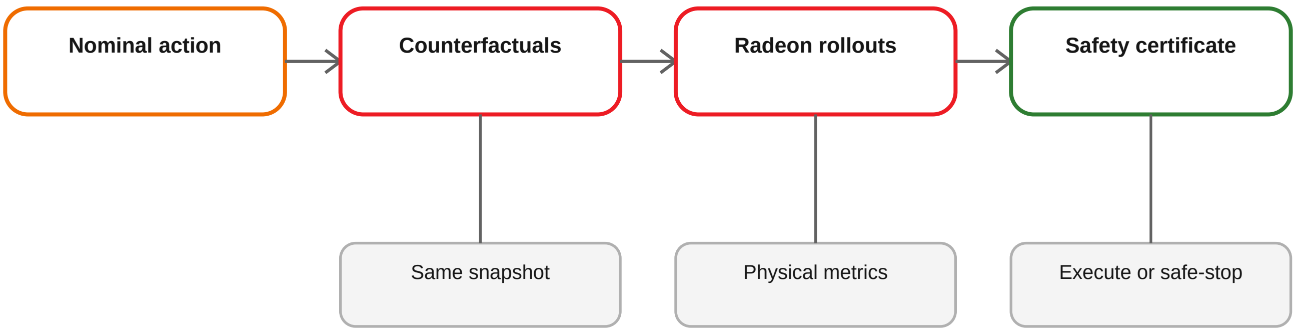
The target application is safety-aware action selection for robot manipulation in clutter. The retained reference task uses a Franka Panda arm to pick a banana, lemon, or plum from a table and move it toward a bowl. A nominal scripted policy can complete many layouts, but its fixed grasp orientation and approach geometry do not explicitly reason about nearby clutter.

GuardianSim addresses a narrower and auditable question:

> *Before executing a policy-proposed grasp, can a simulator evaluate bounded alternatives from the same physical state, reject actions that violate safety margins, and either execute a safer eligible action or stop?*

The project focuses on the decision boundary rather than claiming a new end-to-end robot foundation model. This makes every action choice inspectable and every published metric traceable to a preserved report.

2. System architecture



The implementation is split into four layers:

1. **Simulator-independent decision core.** Candidate models, scoring, recovery logic, serialization, and benchmark validators do not depend on Genesis runtime objects.
2. **Snapshot-safe simulation adapter.** A captured episode state contains robot joints, object poses, and a deterministic fingerprint. Every candidate rollout restores this state before evaluation.
3. **Physical measurement layer.** Rollouts produce reachability, grasp alignment, retained-lift stability, path length, uncertainty, and minimum sampled clutter clearance. Support-surface contact is diagnosed separately from non-target clutter overlap.
4. **Safety-first selector.** Hard gates determine eligibility before utility ranking. The selector never falls back to an unsafe nominal action.

3. Method

3.1 Counterfactual action space

The initial candidate matrix varied grasp yaw and lateral gripper offsets. The final Gate 3.2 action space was expanded after the Gate 3.1 negative result. It contains 18 obstacle-aware candidates:

- yaw angles from -90 degrees to +90 degrees in 22.5-degree increments;
- the nominal target position;
- a 0.025 m retreat direction away from primary clutter;
- a raised 0.14 m approach for retreating candidates.

The nominal action remains represented explicitly, so GuardianSim can retain it when it is already safe.

3.2 Snapshot identity

Comparing candidates from different physical states would invalidate the counterfactual claim. GuardianSim therefore:

- captures one settled episode snapshot;
- fingerprints its robot and object state;
- restores the same snapshot before each candidate rollout;
- validates the fingerprint when resuming a report.

During development, strict resume validation correctly rejected two attempts to append a smoke report from a newly initialized Genesis process whose base snapshot fingerprint differed. Those rejected logs are preserved as audit evidence.

3.3 Physical measurements

For each candidate, GuardianSim measures:

- inverse-kinematics reachability;
- grasp-axis alignment;
- retained object lift and stability;
- sampled path length;
- perception uncertainty;
- minimum AABB separation between distal arm links and named non-target clutter.

A diagnostic identifies the responsible sample, robot link, obstacle, overlap state, and overlap depth. Table support contact is recorded separately and does not determine clutter collision risk.

3.4 Safety eligibility and repeatability

Gate 3.2 freezes all thresholds before the formal run. A candidate is eligible only if its conservative aggregate passes every reachability, stability, and minimum-clearance gate. Candidate confirmation uses repeated rollouts; formal execution then independently executes each strategy three times. An episode is a repeatable safe completion only when all three executions pass.

The selector returns one of four explainable decision types:

- retain an eligible nominal action;
- use a higher-margin alternative;
- replace an unsafe nominal action;
- safe-stop because no action is eligible.

3.5 Post-condition monitoring

Independent execution records whether the gripper closed, object lift was retained, clutter was contacted, the clearance threshold was violated, or the lift became unstable. Failure diagnosis maps these observations to bounded recovery actions. The benchmark does not count a visually plausible but unsafe execution as a success.

4. Training and evaluation data

GuardianSim's primary results do not require a newly trained policy or an external training dataset. The nominal action comes from the retained scripted Franka fruit-picking reference pipeline. The contribution is the counterfactual safety layer and its physical evaluation.

The frozen Gate 3.2 evaluation matrix contains 30 procedurally declared, previously unseen scenarios:

- three target objects: banana, lemon, and plum;
- lateral and radial clutter configurations;
- five deterministic seeds per object/configuration cell;
- three independent physical executions for each strategy and scenario.

The scenario order, thresholds, protocol hash, and matrix hash were frozen before the formal outcomes were inspected:

- protocol SHA-256: `8f23247001e05f39817225ed13f028321fbb9b9c694aaacd5b987fe61ee1fb3c`;
- matrix SHA-256: `69f87994b87f2def788cd944ad75210cdeddeafcaa3d0a3844fef04efca9cb03`.

Gate 3.3 adds engineering breadth tests for pose shift and changed gap/bearing. Its completed 12-scenario prefix is reported separately and is not combined with the primary 30-scenario performance claim.

5. AMD Radeon GPU use

The formal benchmark and evaluator smoke ran on AMD Radeon Cloud using the Genesis `gs.amdgpu` backend.

Recorded formal environment:

- AMD Radeon Cloud Blank OpenCode Workspace;
- one AMD Radeon device, PCI model `0x744b`, GFX architecture `gfx1100`;
- approximately 47.98 GB device memory reported by Genesis;
- Python 3.12.3;
- PyTorch `2.9.1+gitff65f5b`;
- HIP `7.2.53211-e1a6bc5663`;
- Genesis 1.2.3.

The GPU accelerates Genesis scene stepping and the repeated physical counterfactual rollouts. Mean Gate 3.2 planning wall time was 264.95 seconds per scenario for the frozen candidate and confirmation protocol. Mean independent execution time was 9.08 seconds for the baseline and 8.94 seconds for GuardianSim.

The repository includes exact ROCm wheel installation, a dependency lock, an environment-manifest collector, a GPU-required preflight, a bounded three-candidate Genesis smoke, and a complete-source ROCm Dockerfile.

6. Results

6.1 Primary Gate 3.2 result

Metric	Nominal baseline	GuardianSim	Difference
Repeatable safe completion	18/30 (60%)	30/30 (100%)	+40.00 percentage points
Independent safe executions	58/90	90/90	+32 executions
Clutter-contact executions	30	0	-30
Mean sampled clutter clearance	0.023191 m	0.046003 m	+98.36%
Mean retained-lift stability	0.892762	0.905099	+0.012337

GuardianSim decisions comprised 11 higher-margin alternatives, 10 replacements of unsafe nominal actions, and nine eligible nominal fallbacks. No Gate 3.2 scenario required a safe stop.

6.2 Negative evidence and iteration

Gate 3.1 is intentionally retained. It showed that increasing average clearance alone did not improve safe-completion rate. This negative result motivated three Gate 3.2 changes:

- expanded obstacle-aware action geometry;
- an explicit rule forbidding fallback to an unsafe nominal action;
- three independent physical executions for repeatability.

6.3 Breadth evidence and limitation

In the independent 12-scenario Gate 3.3 prefix, GuardianSim executed ten actions safely and safe-stopped twice, with zero clutter contacts or clearance-violating executions. The nominal baseline completed seven safely, made four clutter contacts, and had one clearance violation.

The two stops occurred in lateral lemon and plum gap/bearing cases where no candidate satisfied all frozen hard gates. This is correct fail-safe behavior, but it exposes a real action-space coverage limitation. GuardianSim does not claim universal completion under arbitrary geometry.

7. Innovation and technical contributions

1. **Counterfactual safety layer rather than another task policy.** GuardianSim can wrap a policy-proposed action and reason about alternatives before physical execution.
2. **Snapshot-safe candidate comparison.** Every alternative is evaluated from an identical fingerprinted scene state, and incompatible cross-process resumes are rejected.
3. **Hard eligibility before ranking.** Utility cannot compensate for failed safety requirements; no eligible action results in an explicit stop.
4. **Repeatability-aware evidence.** Formal success requires three independent safe executions rather than one favorable rollout.
5. **Explainable diagnostics and audit trail.** Reports preserve physical metrics, responsible links/obstacles, decision taxonomy, frozen protocol identity, logs, and checksums.
6. **Failure-driven development.** The preserved Gate 3.1 negative result and Gate 3.3 safe-stop limitation constrain the claims and guide future work.

8. Reproducibility and deliverables

Primary deliverables:

- source code and bundled simulation assets;
- `uv.lock` and exact Radeon wheel installer;
- complete-source ROCm Dockerfile;
- machine-readable environment capture;
- one-command source/GPU preflight;
- bounded real Genesis counterfactual smoke;
- immutable Gate 3.2 schema-5 report and validator;
- Gate 3.3 breadth evidence;
- annotated comparison demo and interactive showcase;
- technical report and 3-5 minute video.

The evaluator path is documented in the repository root `REPRODUCIBILITY.md`. The formal report validator must reproduce 30 episodes and the frozen protocol hash before any primary metric is accepted.

9. Limitations and responsible claims

- All evidence comes from Genesis simulation.
- No physical robot was tested.
- Axis-aligned sampled clearance is an engineering proxy, not a formal continuous-time collision proof.
- The evaluated object set is limited to three YCB fruits and declared clutter configurations.
- Planning is not yet real-time.
- Safe stopping protects against unsupported geometry but reduces task completion.
- The reported 30-scenario result applies only to the frozen Gate 3.2 matrix.

Future work should expand continuous grasp geometry, use richer collision distance models, calibrate perception uncertainty from camera observations, reduce planning latency through parallel rollouts, and validate on physical hardware.

10. Team contributions

Member	Contribution
Solo developer - GitHub `@nvm-star-max`	Project direction, system design, implementation, Radeon Cloud experiments, evidence preservation, documentation, and demo production.

AI-assisted development tools were used for implementation and documentation support. The submitting team remains responsible for technical validation, claims, licenses, and competition-rule compliance.

References

1. GuardianSim source and evidence: https://github.com/nvm-star-max/GuardianSim
2. Genesis Embodied AI: https://github.com/Genesis-Embodied-AI/Genesis
3. LeRobot: https://github.com/huggingface/lerobot
4. ROCm documentation: https://rocm.docs.amd.com/
5. Upstream Franka fruit-picking reference: https://github.com/wangxunx/franka_fruit_pick_demo